

ital receivers are now being modelled in oscilloscopes. Bhatia adds, "Equalisation software are used to quickly verify equaliser chip tap values or provide custom tap values to help engineers open even the most tightly closed eyes. De-embedding software are used to combine measurements and models to view simulated scope measurement results at any location in a design. Design models (S-parameters or transfer functions) of interconnects, probes, traces, etc can be imported in the oscilloscope software and the acquired real-time scope data can be transformed to desired measurement locations."

Jitter analysis. Oscilloscopes have in-built jitter measurement and analysis capability. Jitter is the unwanted part of the signal. It degrades the performance of the system and eludes troubleshooting efforts. It is mandatory for high-speed designs and serial data standards to comply with jitter tests.

"Today, total jitter (Tj) measurement is very important for all high-speed serial data. With the capability to analyse the entire record length and higher measurement accuracy, Tj BER at 10^{-12} provides design confidence for reliable data communication," informs Rajendran.

Signal integrity (SI) analysis. A high-end user needs to look at noise-floor and trigger-jitter of a high-bandwidth oscilloscope. "The intrinsic noise of equipment becomes the most significant factor for high-bandwidth measurement. Some applications where you do jitter measurement, the intrinsic jitter of the scope may produce unreliable results," says Bhatia.

Dhruv adds, "Signal integrity analysis mandates for accurate and repeatable measurements, which calls for better 'effective number of bits' specification of a scope at higher bandwidth."

"New high-performance DSOs/MSOs now use reference-class ADCs that have a very low intrinsic noise,"

shares Prabhu.

Complex triggering. Triggering is required for signal characterisation and stabilising repetitive waveforms. Bhatia shares, "A high-end user may look for triggering capabilities of an oscilloscope to trigger on complex signals. Examples of complex triggering include three-stage triggering, zone-qualify triggering and measurement triggering. These enable the user to trigger on highly complex signals."

Price and value proposition

Oscilloscopes have evolved from analogue scopes to high-performance digital oscilloscopes. While the transition from analogue real-time oscilloscopes (AROs) to DSOs continues, in some application areas, AROs will still be used. "As the ARO market shrinks, the ARO price will go up due to an increase in the component price, especially for CRT," explains Huang.

"As key components of DSOs are becoming commercially available, the DSO development no more relies on the expensive application-specific integrated circuit (ASIC) components. Also, with the evolution in IT technology, high-end DSOs are aiming at new applications. The gap between high-end and low-cost DSOs is becoming smaller every day," adds Huang.

The competition among DSO suppliers will highly rely on better and cheaper new products. "The differentiation of banner specifications is becoming insignificant, as the cost of key components is the same for all the suppliers. So it's all about value proposition—like the functional capabilities offered by different suppliers at a given cost," says Huang.

Scope for improvement

Oscilloscopes have come a long way in terms of functionality, cost, size and variety of integration. These are available for as low as Rs 5000, while the high-end oscilloscopes have bandwidth as high as 30 GHz (real-time) with 80GS/s sampling rates. Oscilloscopes are available in all form fac-

tors—benchtop, handheld, portable and PC-based small boxes dedicated to different work environments. Still, there is a scope for improvement, not just in the equipment but in the usage of equipment too.

"From early valve versions of oscilloscopes, designs have updated to transistors, ICs and the latest embedded technology. Perhaps, the nanotechnology may open new frontiers," adds Prabhu.

Multigrid display utilises the full dynamic range of ADCs for each trace

"All the hobbyists, budding engineers, professionals, designers and test engineers use oscilloscopes, but not the full functionalities," says Rajendran. "The average use is 20-30 per cent of the features, but there are so many useful tools in the scope that can bail them out of troubled waters."

Is it because they do not need all the features? Or because they don't know how to use the oscilloscope?

Anand Bhushan, managing director of Bhushan & Bhushan, explains, "Education requires a basic oscilloscope. Engineers need to understand the concepts of triggering, sampling and measurement of frequency, amplitude, time period, etc. Oscilloscopes with automated measurements are useful in the industry to save time. These give ready results but do not help the budding engineers in learning the underlying processes leading to those results. Black boxes don't educate!"

Test and measurement manufacturers provide user manual, hands-on training and demonstrations to the academia and the industry so that they can explore all the features of a scope. Manuals come handy when you use an oscilloscope for more than viewing a waveform. Following the buying guidelines and reviews on the Internet will help you buy the right scope for your application. ●

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**Many a times,
high-end scopes have
DSP-boosted bandwidth**